CSS Combinators, Pseudo-Classes & Pseudo-Elements

A structured classroom guide to mastering CSS selectors — from element relationships to state-based styling and generated content.





PART 1

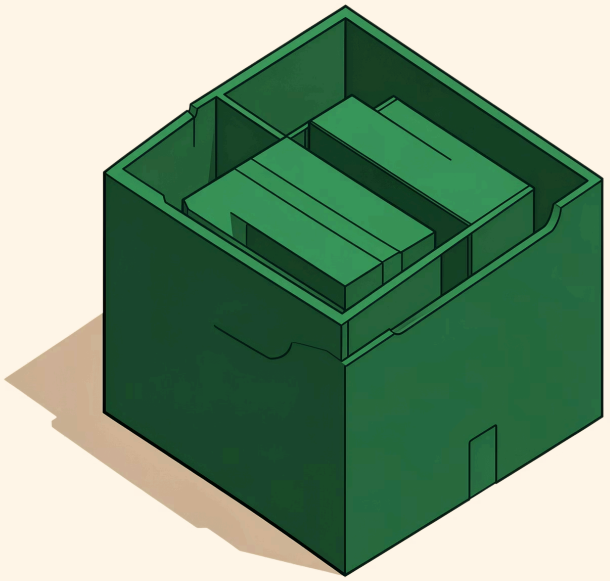
CSS Combinators

How elements relate to each other in the document tree — and how to target them precisely.

What Are Combinators?

Combinators define **relationships between elements** — they let you select elements based on their position or connection to other elements in the HTML tree.

Combinator	Symbol	Meaning
Descendant	`A B`	Any matching element inside A
Child	`A > B`	Direct child of A only
Adjacent Sibling	`A + B`	Immediately following sibling
General Sibling	`A ~ B`	All matching following siblings



Descendant & Child Combinators

Descendant `div p`

Selects all `<p>` anywhere inside `<div>`, at any nesting level.

Child `div > p`

Selects only `<p>` that are *direct* children of `<div>`; nested paragraphs are skipped.

Memory trick: Space = anywhere inside · `>` = directly inside

Sibling Combinators

Adjacent Sibling h2 + p

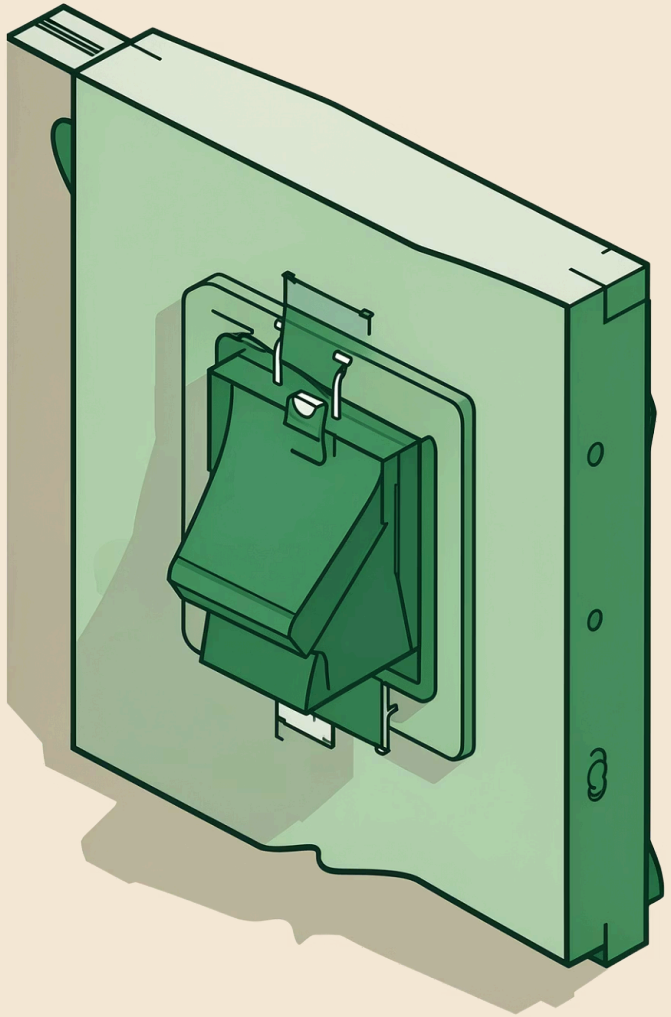
Selects the *one* paragraph that immediately follows an <h2>. Only the very next sibling qualifies.



General Sibling h2 ~ p

Selects *all* paragraphs that are siblings of <h2> and appear after it in the DOM.

Memory trick: + = next one · ~ = all following ones

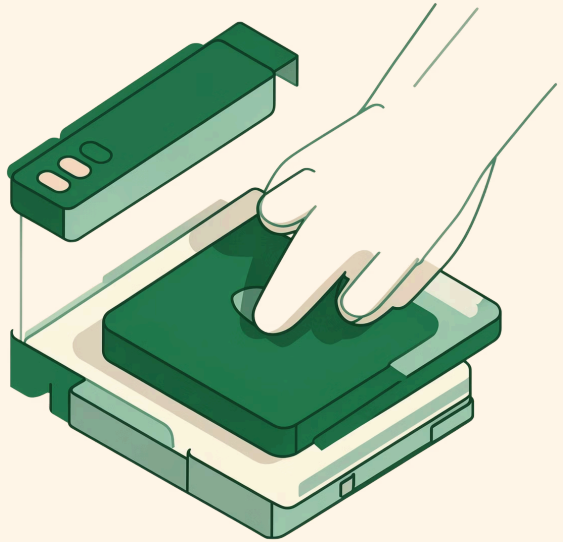


PART 2

Pseudo-Classes

Select elements based on their state, condition, or position — without adding extra classes to your HTML.

Pseudo-Classes: State & Interaction



A pseudo-class selects an element based on its **state, condition, or user interaction** — syntax: `:keyword`

Link order (LVHA): :link → :visited → :hover → :active

`:hover`

Mouse is over the element — buttons, links, cards.

`:active`

Element is being clicked or pressed.

`:focus`

Element has keyboard or click focus — inputs, form controls.

Structural Pseudo-Classes

:first-child / :last-child

Selects the first or last child element of a parent container.

:nth-child(n)

Selects by position. Supports even, odd, or a formula like 2n+1.

:nth-of-type(n)

Like :nth-child but counts only elements of the same tag type.

Key difference: :nth-child counts all siblings · :nth-of-type counts only matching tags



Form & Utility Pseudo-Classes

→ `:checked`

Checked checkbox or radio button.

→ `:valid` / `:invalid`

Input passes or fails validation rules.

→ `:disabled` / `:enabled`

Disabled or enabled form controls.

→ `:not(.class)`

Selects everything *except* the specified selector.

→ `:required` / `:optional`

Required or optional input fields.

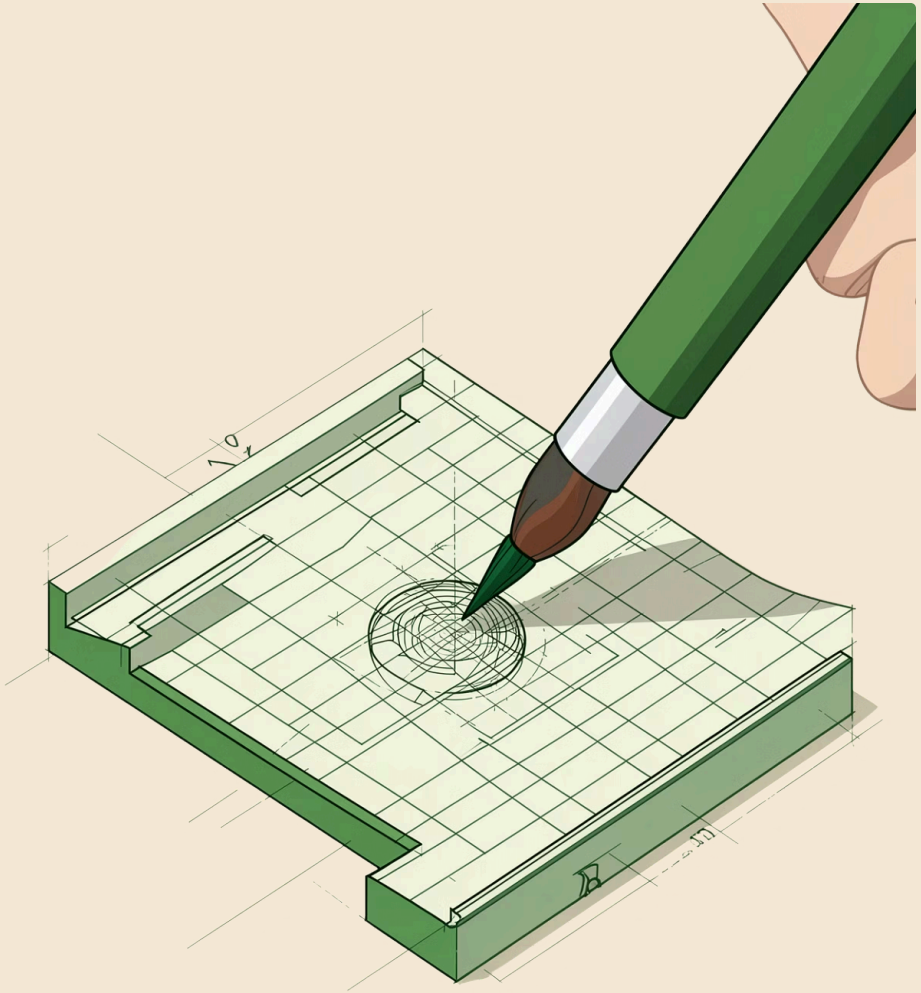
→ `:root`

The `<html>` element — used for CSS custom properties.

PART 3

Pseudo-Elements

Target specific parts of an element or inject generated content — without touching your HTML markup.





Pseudo-Elements: Styling Parts of Elements

A pseudo-element targets a **specific part** of an element or inserts generated content — syntax: `::keyword`

Common uses: icons, decorative shapes, labels, arrows, badges.

`::before`

Inserts content *before* the element's content. Requires the `content:` property.

`::after`

Inserts content *after* the element's content. Same rules apply.

Text & Form Pseudo-Elements



::first-letter

Styles the first letter — classic drop caps in articles.



::first-line

Styles the first line of a text block.



::selection

Styles text highlighted by the user.



::placeholder

Styles placeholder text inside inputs.



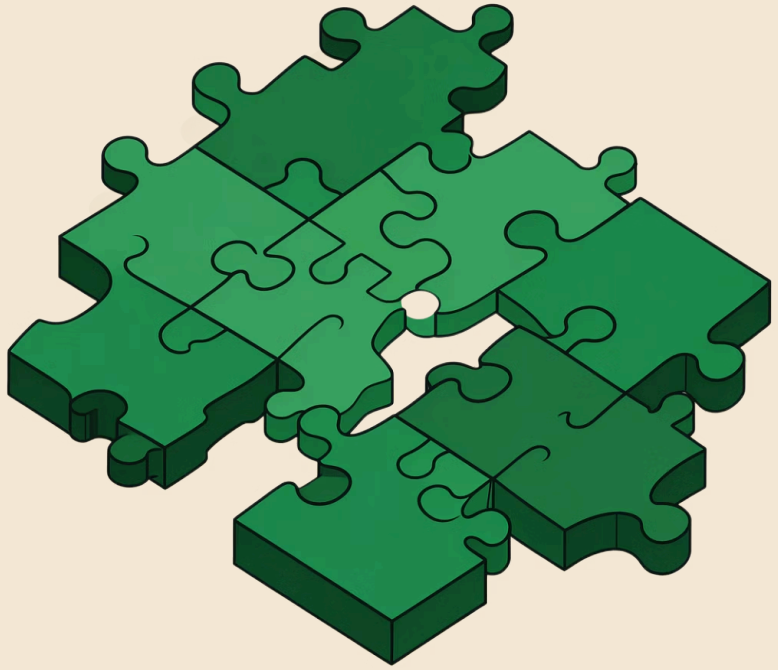
::marker

Styles the bullet or number of list items.

Pseudo-Classes vs Pseudo-Elements

One colon or two? The difference matters — here's how to keep them straight:

	Pseudo-Class	Pseudo-Element
Syntax	Single colon `:`	Double colon `::`
Targets	Element state or condition	A part of an element
Examples	`:hover`, `:focus`, `:nth-child()`	`::before`, `::after`, `::first-letter`
Think...	What state is this element in?	Which part do I want to style?



Combining It All

01

Combinator + Pseudo-Class

`nav a:hover` — links inside `<nav>` on hover.

02

Child + Structural

`.card > p:first-child` — first direct paragraph inside `.card`.

03

Pseudo-Element on a Class

`.card::before { content: "NEW"; }` — adds a badge label automatically.

Combining selectors is where real CSS power comes from.



🧑‍💻 Lab: Course Listing Page

Combinators

.card p · .card > h3 · h3 + p · h3 ~ p

Pseudo-Classes

:hover · :focus · :nth-child(even) · :disabled · :valid · :invalid

Pseudo-Elements

::before (icon) · ::first-letter (drop cap) · ::placeholder · ::marker

🏆 Challenge

- Style the **first card differently** using `:first-child`
- Apply **alternating backgrounds** with `:nth-child(even)`
- Add a **"NEW" badge** via `::before` on featured cards
- Highlight **invalid inputs** in red using `:invalid`